



Challenge 2 - Bundesliga
AWS World Sports Innovation Cup 2026

Unlock the Power of 3D Football Data

1. Background

In recent years, the Bundesliga has taken a pioneering step in sports analytics by transitioning from traditional center of mass tracking to advanced 3D skeletal data collection. This leap forward means that, instead of merely tracking the general position of players on the pitch, we now capture detailed, frame-by-frame information about the movement and posture of every player's body. With over 141 million data points per match collected, a world of possibilities for deeper insights into the game are opened. However, despite this technological advancement, the current use of this data is limited: it primarily powers automated event detection, such as identifying passes, shots, or tackles. The true potential of 3D skeletal data remains largely untapped. Imagine being able to quantify player agility, peripheral scanning abilities or tactical awareness in real time, or developing entirely new ways for fans to engage with the game through immersive visualizations or interactive statistics.

2. Challenge

The challenge invites you to explore this cutting-edge dataset of roughly 705 million data points (= five Bundesliga matches), you have the opportunity to:

1. Develop innovative Key Performance Indicators (KPIs) that go beyond what's possible with traditional data.
2. Create new engagement solutions for Bundesliga channels, partners, or clubs, enhancing the fan experience and/or coaching decisions.
3. Benchmark and adapt best practices from other sports or industries that leverage 3D data.

Your creativity and analytical skills could help define the next generation of football statistics, such as the Bundesliga Match Facts, and data applications, making a real impact on how the game is played, watched, and understood. With the Bundesliga's move to 3D skeletal data, the field is wide open, now it's your turn to lead the way.

3. How to access the AWS environment and the data

You will receive a separate document **ISB-UserGuide.pdf** with instruction on how to access the AWS Account and the challenge data.

4. Deliverables

A single zip-folder (the name must be your team's name, e.g. **MyTeamName.zip**) containing:

- i. **github_link.txt** - A text file containing the URL to your Github Repo. The repository must include your source code (.ipynb, scripts, etc.) and a README.md with clear instructions on how to execute your code and reproduce your results. If you want to keep the repo private, please invite MoellerO to your repo. **Do not upload any hackathon data to the repository.**
- ii. **presentation_video.mp4** - A short video where you can present a demo of something you build or explain your newly computed KPI (e.g. using powerpoint). **Max. 3mins. Low resolution (<720p).**
- iii. **executive_summary.pdf** – Up to five-slide PowerPoint summarizing your solution and displaying the main output from your approach. Note: We value creative problem-solving and learning from failures
- iv. **(Optional) prfaq.pdf** — A [PRFAQ document](#) (PDF format) where you can go into more detail about what you did and why

5. How to Submit

- i. Create your zip folder (MyTeamName.zip) containing the four items listed above. Keep the total size small.
- ii. Use [this file request link](#). You may have to create a free box account.
- iii. Upload your zip folder.
- iv. If you need to resubmit, upload a new zip with the same team's name and add a version suffix (e.g., MyTeamName _v2.zip). Please avoid resubmitting many times.

6. Questions?

- i. Reach us on discord: <https://discord.gg/EBYZNDbwzp>